Title :- Network Devices.

Date of Class:-

Date of Home Practice :-

SUREN NETWORK
PLAY WITH TECHNOLOGIES

SNCNA

SUREN NETWORK CERTIFIED NETWORK ASSOCIATE

DEVICE TEST

of

HUB, SWITCH & ROUTER behaviours.

1, Explain behaviour of hub device

⇒ Hub is a network device. It is a Layer 1 device.

⇒ If we want send data b/w devices first we have form N/w.

⇒ Hub is a network which will send data to devices in same network.

⇒ If data is sending b/w devices first we need to find destination MAC. Here Hub will broadcast for destination MAC. After finding a destination MAC, from source sender will send data to Hub then hub will send to destination MAC.

2, Explain behaviour of switch device

⇒ Switch is a Network device. It is a data link (L2) device.

⇒ Switch is more intelligent device then the Hub.

⇒ A switch uses "MAC addresses" to decide where to forward Ethernet frames.

⇒ It examines the Source MAC address. It learns/stores that MAC address in its MAC address table

⇒ It examines the destination MAC address.

⇒ If the destination MAC is known, It forwards the frame only through the appropriate port.

⇒ If the destination MAC is Unknown, It floods the ~~frame~~ out.

⇒ Broadcast frames are also flooded within the VLAN.

3, Explain behaviour of router device

⇒ A Router is a Layer 3 (Network Layer) device.

⇒ A Router is a network device used to Connect b/w different N/w.

⇒ A router uses "IP addresses" and its "routing table" to determine where packets should go.

⇒ Examines the destination IP address.

⇒ Looks up the destination network in its routing table,

⇒ Select the best route.

Easy way to remember:

Hub → Send to everybody

Switch → Send using MAC Address Table.

Router → Send using IP address & Routing table.

For CCNA Notes:

HUB → Repeat

SWITCH → Forward

ROUTER → Route.